



Jashore University of Science and Technology
Department of Computer Science and Engineering

Semester Final Examination-2023
2nd Year 2nd Semester, Session: 2020 - 2021

Course Code: CSE 2201

Time: 3.00 hours

Course Title: Data Communication

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. a) Define data communication systems. Briefly describe the major components of a data communication system. 04
- b) "A star topology is less expensive than a mesh topology" - Is it true or false? Justify your answer. 03
- c) Define protocols and standards in data communication. Briefly describe the key elements of a protocol. 05
2. a) Briefly describe different types of addressing used in computer network. 03
- b) What are the layers in the OSI network model? Write the functions of the data link layer. 03
- c) If a periodic signal is decomposed into five sine waves with frequencies of 200Hz, 300Hz, 400Hz, 700Hz, 800Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude to 5V. 04
- d) Assume we need to download a text document at the rate of 50 pages per minute in a channel. What is the required bit rate of the channel? 02
3. a) Define transmission impairment. Briefly describe the signal attenuation with an illustration diagram. 04
- b) Suppose a signal travels through a transmission medium and its power is reduced to one-half. Compute the signal attenuation in dB. 02
- c) What are the propagation time and the transmission time for a 2.5-kbyte message (an e-mail) if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s. 04
- d) What does the Nyquist theorem have to do with communications? 02
4. a) Define unguided transmission media. Briefly describe different types of propagation modes in optical fiber communication. 06
- b) Define connectors. Briefly describe different types of connectors used in coaxial cable connections. 04
- c) A network with bandwidth of 10Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network? 02

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5. a) Compare Polar NRZ-L and NRZ-I schemas of line coding. 03
- b) Briefly describe the Manchester and differential Manchester encoding schemes. 05
- c) A signal is carrying data in which one data element is encoded as one signal element. If the bit rate is 100 kbps, what is the average value of the baud rate if $c = 0.5$? 02
- d) A system is using NRZ-I to transfer 1-Mbps data. What are the average signal rate and minimum bandwidth? 02
6. a) Define modulation. Describe the process of amplitude modulation. 05
- b) Describe the QPSK method of digital to analog conversion. 05
- c) We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What are the carrier frequency and the bit rate if we modulated our data by using ASK with $d = 1$? 02
7. a) What is the Hamming distance for each of the following code words? 04
- d (10100, 10000)
 - d (10101, 10000)
 - d (11111, 11111)
 - d (0100, 0001)
- b) Apply the following operations on the corresponding polynomials: 04
- $(x^3 + x^1 + x + 1) + (x^4 + x^2 + x + 1)$
 - $(x^3 + x^1 + x + 1) - (x^4 + x^1 + x + 1)$
 - $(x^3 + x^1) \times (x^4 + x^2 + x + 1)$
 - $(x^3 + x^2 + x + 1) / (x^2 + 1)$
- c) Why Switching is needed? Make comparative analysis among circuit switched, packet switched and message switched networks. 04
8. a) Differentiate between Pure aloha with slotted aloha. 04
- b) Draw the flowchart of CSMA/CD. Explain the working principles of CSMA/CD. 05
- c) Write short notes on: 03
- Polling
 - Token passing
 - Channelization



Department of Computer Science and Engineering

Jashore University of Science and Technology

Semester Final Examination-2023

B.Sc. (Engg.) 2nd Year 2nd Semester, Session: 2020-21

Course Code: CSE 2203

Time: 3.00 hours

Course Title: Database Management System

Marks: 72

[N.B. Answer any 6 set questions from the following 8 set of questions. The figures in the right margin indicate full marks.]

1. (a) Define Database Management System (DBMS). Mention the advantages of DBMS over the conventional file processing system. 3
(b) Write down any two major responsibilities of a database administrator. 2
(c) "All candidate keys are super keys, but all super keys are not candidate keys" – justify this statement. 3
(d) Explain the term distributed database management system (DDBMS). Mention the issues to be considered in the design of DDBMS. 4
2. (a) From the following, identify DDL, DML, Logical design, Physical design SQLs 6
 - I. `CREATE INDEX idx_lastname ON Persons (LastName);`
 - II. `DROP INDEX index_name ON table_name;`
 - III. `DROP TABLE table_name;`
 - IV. `SELECT CustomerName, ContactName INTO CustomersBackup2017 FROM Customers;`
(b) "A relational data model relates the data instances" – Justify 2
(c) Explain query processor 4
3. (a) Consider the following relational scheme of an employee database 6*2=12
Employee (employee_name, street, city)
Works (employee_name, company_name, salary)
Company (company_name, city)
Manages (employee_name, manager_name)

You are required to write Structured Query Language to:
 - I. Find the names of all employees who work for First Bank Corporation.
 - II. Find all employees in the database who live in the same cities as the companies for which they work.
 - III. Find all employees in the database who live in the same cities and on the same streets as do their managers.
 - IV. Find all employees who earn more than the average salary of all employees of their company.
 - V. Find the company that has the smallest payroll.
 - VI. Display all managers of First Bank Corporation a 10 percent raise in salary. (
4. (a) Consider the following schema 8
Student (RollNo, FirstName, EnglishMarks, PhysicsMarks, ChemistryMarks, MathsMarks, TotalMarks);

You are required to:
Create a Trigger for each row to calculate and set the value of *TotalMarks* while inserting value of other columns. Please note that *TotalMarks* = *EnglishMarks* + *PhysicsMarks* + *ChemistryMarks* + *MathsMarks*
(b) Define functional dependency. Describe different types of functional dependencies with proper example. 4
5. Design a database for an airline. The database must keep track of customers and their reservations flights and their status, seat assignments on individual flights, and the schedule and routing of future flights 4+4+4

Your design should include:
 - I. An E-R diagram.
 - II. A set of relational schemas (derived from E-R diagram)
 - III. A list of constraints, including primary-key and foreign-key constraints
6. (a) Normalize the following schema with given constraints, to 4NF 8
books (accession_no, isbn, title, author, publisher)
users (user_id, name, dept_id, dept_name)
accession_no → isbn

$isbn \rightarrow title$
 $isbn \rightarrow publisher$
 $isbn \rightarrow author$
 $user_id \rightarrow name$
 $user_id \rightarrow dept_id$
 $dept_id \rightarrow dept_name$

(b) Explain ACID properties.

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4

7. (a) "Consistency of the Database must be preserved even with the expense of Concurrency" - Justify with the following schedule

6

T1	T2
Read(A)	
$A \leftarrow A - 50$	
Write(A)	Read(A)
Read(B)	Read(B)
$B \leftarrow B + 10$	$Z \leftarrow A + B$
Write(B)	Write(Z)
Commit	Commit

(b) "Cascading Rollback may waste massive amount of resources" - explain with respect to the following schedule

6

T1	T2	T3
Read(A)		
Read(B)		
Write(A)		
	Read(A)	
	Read(B)	
		Read(A)
Abort		

8. (a) Explain the evolution of locking protocol based on the following schedule

6

T1	T2	Concurrency Control Manager
Lock-X(B)		
Read(B)		Grant-X(B, T1)
$B_1 = B - 50$		
Write(B)		
Unlock(B)		
	Lock-B(A)	Grant-S(A, T2)
	Read(A)	
	Unlock(A)	
	Lock-S(B)	Grant-S(B, T2)
	Read(B)	
	Unlock(B)	
	Display(A+B)	
Lock-X(A)		Grant-X(A, T1)
Read(A)		
$A_1 = A + 50$		
Write(A)		
Unlock(A)		

(b) "Locking protocols may produce deadlock. But deadlock is better than inconsistency" - justify with the following what for graph, P1, P2, P3, P4 and P5 are transactions

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Department of Computer Science and Engineering

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Semester Final Examination-2023

B.Sc. (Engg.) 2nd Year 2nd Semester, Session: 2020-21

Course Code: CSE 2205

Time: 3.00 hours

Course Title: Digital System Design

Marks: 72

[N.B. Answer any 06 questions from the following 08 sets of questions. The figures in the right margin indicate full marks.]

- 1 a) How many parts are there of a statement in the register-transfer language? Describe their role in register transfer. 2
- b) Describe the types of microoperations most often encountered in digital systems. 4
- c) Illustrate the hardware implementation of the following statement- 6
 - $xyT_0 + T_1 : A \leftarrow A + B$
 - $xyT_2 : A \leftarrow A + 1$
- 2 a) Perform the arithmetic operations $(+57) + (-39)$ and $(+57) - (-39)$ using 6
 - i. Sign-2's complement
 - ii. Sign-10's complement
- b) Explain the circuit of addition of sign-2's complement numbers with overflow. 4
- c) Differentiate between Logic microoperation and Shift microoperation. 2
- 3 a) Demonstrate the logic diagram of the arithmetic circuit for 2-bit operation. 4
- b) Describe the use status register in the digital system and the role of different flag bits with examples. 4
- c) Design a 4-bit combinational logic shifter with serial input. 4
- 4 a) Write short notes on- 4
 - i. Hard-wired control
 - ii. Microprogram sequencer.
- b) "One flip-flop per state method is called the ring-counter controller" Justify the statement with proper reasoning and explanation. 4
- c) Discuss the usage of the next address generator in the microprogram control method in your own words. 4
- 5 a) Write the list of the sequence of instructions for the computer that will 4
 - i. Reset flip-flop E
 - ii. Set flip-flop F
 - iii. Set flip-flop S
 - iv. Set bit N₉
- b) Generate the list of register-transfer statements for the execution of the following 6

iii) both times.

- b) One shot is fired from each of the three guns. E_1, E_2, E_3 denote the events that the target is hit by the first, second and third guns respectively. If $P(E_1) = 0.5, P(E_2) = 0.6$ and $P(E_3) = 0.8$ and E_1, E_2, E_3 are independent events. find the probability that 0.4

instructions. Assume that the computer does not have an F flip-flop, but that the sequence register has 16 timing variables t_0 through t_{15} . The G register must be cleared when the execution of the instruction is completed.

Symbol	Hex code	Description	Function
SBA	8 m	Subtract from A	$A \leftarrow A - M$
ADM	9 m	Add to memory	$M \leftarrow A + M$

- c) Differentiate between direct transfer and transfer with handshaking in a parallel peripheral interface. 2
- 6 a) Write short notes on- 4
- Addressing modes
 - Interrupt
 - BSB
 - Bus buffer
- b) What will the accumulator register A and the status bits C (carry), S (sign), Z (zero), and V (overflow) contain after each of the following instructions? The initial value of register A in each case is $(83)_{16}$. 8
- CMA (Complement A) $A \leftarrow \bar{A}$
 - RLC (Rotate A left with carry) $A \leftarrow clc A$
 - RRC (Rotate A right with carry) $A \leftarrow crc A$
 - Exclusive NOR to the accumulator itself
- 7 a) A digital computer has a memory unit with 24 bits per word. The instructions set consists of 190 different operations. all instructions have an operation code part(opcode) and an address part (allowing for only one address) Each instruction is stored in one word of memory. 4
- How many bits are needed for the opcode?
 - How many bits are left for the address part of the instruction?
 - What is the maximum allowable size of memory?
 - What is the largest signed binary number to be stored in one word of memory?
- b) A digital system has three registers AR, BR, and PR. Three flip flops provide the control variables for the system: S which is set by an external Start signal to start the operation, and F and R are used in sequencing the micro-operations when the system is in operation. A fourth flip flop D is set by when the operation is completed. The operation of the digital system is described by the following register transfer statements: 8
- $$S: PR \leftarrow 0, S \leftarrow 0, D \leftarrow 0, F \leftarrow 1$$
- $$F: F \leftarrow 0, \text{if } (AR = 0) \text{ then } (D \leftarrow 1) \text{ else } (R \leftarrow 1)$$
- $$R: PR \leftarrow PR + BR, AR \leftarrow AR - 1, R \leftarrow 0, F \leftarrow 1$$
- Draw a block diagram showing the hardware implementation of the system. Include a start input set flip-flop S and a done output from flip-flop D.
- 8 a) Explain the operation of diode ROM. 3
- b) Design a PLA control system for binary multiplier. To derive the system, draw necessary flow chart, control state diagram, sequence of micro-operations and state table. 9



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Semester Final Examination-2023
 2nd Year 2nd Semester, Session: 2020 - 2021

Course Code: MATH 2201

Time: 3.00 hours

Course Title: Probability and Statistics for Engineers

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. a) What do mean by Statistics according to Statistical methods? Describe the importance of statistics in different sectors. 1+3

- b) An incomplete frequency distribution is given as follows: 1+3

Variable	Frequency	Variable	Frequency
10 - 20	12	50 - 60	?
20 - 30	30	60 - 70	25
30 - 40	?	70 - 80	18
40 - 50	65	Total	229

Given the median value is 46, determine the missing frequencies, using the median formula.

- c) Show that t - distribution tends to normal distribution, if the degrees of freedom tend to infinity. 04
2. a) A two-dimensional r.v. (X, Y) have a bivariate distribution given by: $p(X=x, Y=y) = (x^2+y)/32$, for $x = 0, 1, 2, 3$ and $y = 0, 1$. Find the marginal distribution of X and Y . 04
- b) Three newspapers A, B, and C are published in a certain city. It is estimated from a survey that of the adult population: 20% read A, 13% read B, 16% read C, 9% read both A and B, 5% read both A and C, 4% read both B and C, 2% read all three. Find what percentage read at least one of the papers? 05
- c) What is conditional probability? If $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.3$, then find $P(A|B)$ and $P(B|A)$. 03

3. a) Define Kurtosis and Dispersion. Write the characteristics of an Ideal measure of Central Tendency. 2+2
- b) The median and mode of the following wage distribution are known to be Tk. 3,350 and Tk. 3,400 respectively. Find the values of f_3 , f_4 and f_5 : 04

Wages (in Tk.)	No. of Employee	Wages (in Tk.)	No. of Employees
0 - 1,000	4	4,000 - 5,000	f_5
1,000 - 2,000	16	5,000 - 6,000	6
2,000 - 3,000	f_3	6,000 - 7,000	4
3,000 - 4,000	f_4	Total	230

- c) If $\bar{x}$ and s denote respectively the mean and standard deviation of a set of n observations, show that $\bar{x}\sqrt{(n-1)} \geq s$. 04
4. a) Define probability. Among the digits 1, 2, 3, 4, 5 at first one is chosen and then a second selection is made among the remaining four digits. Assuming that all twenty possible outcomes have equal probabilities, find the probability that an odd digit will be selected
- the first time,
 - the second time, and
 - both times.
- b) One shot is fired from each of the three guns. E_1, E_2, E_3 denote the events that the target is hit by the first, second and third guns respectively. If $P(E_1) = 0.5$, $P(E_2) = 0.6$ and $P(E_3) = 0.8$ and E_1, E_2, E_3 are independent events, find the probability that 04

(i) exactly one hit is registered, and (ii) at least two hits are registered.

- c) An urn contains 6 white, 4 red and 9 black balls. If 3 balls are drawn at random, find the probability that 04

- Two of the balls drawn are red,
- One is of each color,
- None is white, and
- At least one is black

5. a) Define null hypothesis and alternative hypothesis. 03

- b) Distinguish between Type I error and Type II error. 03

- c) How can you test differences of variances of two populations? Is it possible to test variances using t-test? Which statistical test is applicable for test of independence? 03

- d) The average daily intakes of dairy products for men (sample size=50, sample mean=756 and sample 2% standard deviation=35) and women (sample size=50, sample mean=762 and sample standard deviation=30). Is there a difference in the average daily intakes of dairy products for men versus women? (At 5% level of significance, the critical region is $z > 1.96$ or $z < -1.96$) 03

6. a) If X and Y are independent show that they are uncorrelated. 03

- b) Calculate the correlation coefficient for the following heights (inches) of fathers (X) and their sons (Y): 03

X:	65	66	67	67	68	69	70	72
Y:	67	68	65	68	72	72	69	71

- c) Ten competitors in a musical test were ranked by the three judges A, B, and C in the following order: 03

Ranks by A:	1	6	5	10	3	2	4	9	7	8
Ranks by B:	3	5	8	4	7	10	2	1	6	9
Ranks by C:	6	4	9	8	1	2	3	10	5	7

Using rank correlation method, discuss which pair of judges has the nearest approach to common linking in music.

- d) Obtain the equations of two lines of regression for the following data. Also obtain the estimate of X for Y = 70. 03

X:	55	66	67	67	68	69	70	72
Y:	67	68	65	65	72	72	69	71

7. a) A manufacturer, who produces medicine bottles, find that 0.1% of the bottles are defective. The bottles are packed in boxes containing 500 bottles. A drug manufacturer buys 100 boxes from the producer of bottles. Using Poisson distribution, find how many boxes will contain: 03

- No defective, and
- At least two defectives.

- b) Define Normal Distribution. Write the Characteristics of normal distribution and normal probability curve. 1+2

- c) The probability of a man hitting a target is $\frac{2}{3}$. 04

(i) If he fires 7 times what is the probability of his hitting the target at least twice?

(ii) How many times must he fire so that the probability of his hitting the target at least once is greater than $\frac{2}{3}$?

d) With the usual notation, find p for a binomial variate X , if $n = 6$ and $9P(X = 4) = P(X = 2)$. 02

8. a) From a city population, the probability of selecting (i) a male or a smoker is $7/10$, (ii) a male smoker is $2/5$, and (iii) a male, if a smoker is already selected is $2/3$. Find the probability of selecting 03

(a) a non-smoker,

(b) a male, and

(c) a smoker, if a male is first selected.

b) An experiment consists of three independent tosses of a fair coin. Let X = The number of heads, Y = the number of head runs, Z = the length of head runs. A head runs begin defined as consecutive occurrence of at least two heads, and its length then begin the number of heads occurring together in three tosses of coin. 05

Find the probability function of (i) X , (ii) Y , (iii) Z , (iv) $X+Y$, and (v) XY and construct probability tables and draw their probability charts.

c) The diameter, say X , of an electric cable, is measured to be a continuous r.v with p.d.f.: 04
 $f(x) = 6x(1-x)$, $0 \leq x \leq 1$

i) Check that the above is a p.d.f..

ii) Obtain an expression for the c.d.f. of X .

iii) Compute $P(X \leq 1/2 \mid 1/3 \leq X \leq 2/3)$, and .

iv) Determine the number K such that $P(X < k) = P(X > k)$.



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Semester Final Examination-2023
2nd Year 2nd Semester, Session: 2020 - 2021

Course Code: MATH 2203

Time: 3.00 hours

Course Title: Linear Algebra and Fourier Analysis

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. a) Define Symmetric, Skew Symmetric and Idempotent matrices with examples. 06

- b) Prove the matrix $A = \begin{bmatrix} \cos a & -\sin a & 0 \\ \sin a & \cos a & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is orthogonal. 06

2. a) Solve the following system of linear equations using the Gauss-Jordan elimination method: 04

$$2x + 3y + 5z = 10$$

$$4x - 2y + 3z = 8$$

$$3x + 2y + 4z = 18$$

- b) Explain the minor, cofactor and rank of a matrix. Identify each of them for the following matrix 2+3

$$A = \begin{bmatrix} 1 & 3 & -2 & -1 \\ 2 & 6 & -4 & -2 \\ 1 & 3 & -2 & 1 \\ 2 & 6 & 1 & -1 \end{bmatrix}$$

- c) A skew-symmetric matrix satisfies $A^T = -A$, as in 03

$$A = \begin{pmatrix} 0 & 1 & 2 \\ -1 & 0 & 3 \\ -2 & -3 & 0 \end{pmatrix}$$

Briefly discuss why the $\det(A^T)$ is always equal to $\det(A)$?

3. a) State the Cayley Hamilton theorem and verify the theorem for the matrix 08

$$B = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

- b) Explain the singular matrix and its properties. Show the following matrix is singular 04

$$A = \begin{pmatrix} 1 & \sin a & \cos a \\ \sin a & \cos a & -\sin a \\ \cos a & -\sin a & 0 \end{pmatrix}$$

4. a) Explain the RREF with its properties. Why it is important? 3+1

b) Derive the RREF of the following matrix with clearly demonstration of the steps

08

$$\begin{bmatrix} 2 & 5 & 7 & 6 \\ 1 & 5 & 2 & 4 \\ 2 & 4 & 6 & 2 \\ 4 & 3 & 6 & 8 \end{bmatrix}$$

5. a) Describe the Cauchy-Schwarz inequality for 3d vectors. 04

b) Find a line in $\mathbf{R}^3$ through two points A (-2, 10, 0) and B (4, 2, 2). Consider the direction vector AB (6, -8, 2). 04

c) Derive the vector projection theorem. Find the projection of the vector A (4,2,1) on the vector B (5, -3, 3) 2+2

6. a) State and prove the Convolution theorem of Fourier Transformation. 06

b) Identify the Eigenvectors and Eigenvalues for the following matrix 06

$$A = \begin{pmatrix} 1 & 3 & 1 \\ 3 & 2 & 0 \\ 4 & 0 & 1 \end{pmatrix}$$

7. a) Describe the Gibbs Phenomenon and Parseval's Theorem with examples. 3+3

b) How to determine the odd and even functions of Fourier series. Discuss the coefficients for each type of the functions. 3+3

8. a) Derive the Fourier series for the function $f(x) = x \cos \theta$ for the period $-\pi$ to π . 09

b) Prove that $\text{curl grad } \phi = 0$. 03



Department of Computer Science and Engineering

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Semester Final Examination-2023

B.Sc. (Engg.) 2nd Year 2nd Semester, Session: 2020-21

Course Code: HUM 2201

Course Title: Business Psychology

Time: 3.00 hours

Marks: 72

[N.B. Answer any 06 questions from the following 08 sets of questions. The figures in the right margin indicate full marks.]

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- | | | |
|---|---|----|
| 1 | (a) Describe brief history of Business, Industrial and Organizational psychology. | 12 |
| 2 | (a) What is personality? How personality can be shaped? | 06 |
| | (b) Discuss issues of Rewards vs. Punishments? Which one is better for business related goal achieving? | 06 |
| 3 | (a) Is an employee predisposed to be motivated? Describe the individual difference traits that are most related to work motivation. | 12 |
| 4 | (a) What individual differences affect Job Satisfaction? | 06 |
| | (b) How digital transformations have affected Business Psychology? | 06 |
| 5 | (a) Describe types of Organizational Communication. | 12 |
| 6 | (a) What is Group? How leadership is important for a group? | 06 |
| | (b) Which one is better in terms of Business Opportunity: Group Marketing or Mass Marketing? Argue. | 06 |
| 7 | (a) How does business psychology play role in decision making? | 06 |
| | (b) Analyze the impact of corporate culture on personality formation. | 06 |
| 8 | (a) What are the factors affecting Group Performance? | 09 |
| | (b) What type of Group(s) can be exploited easily for Business purposes? Argue. | 03 |